

## Abstract

Pacific herring (*Clupea pallasii*) spawning creates a brief, spatially concentrated coastal resource pulse. Whether this pulse leaves a temporally localized, taxon-specific signature in community-science observations remains unresolved. We tested whether herring-associated birds were reported more often and in higher positive counts near spawn, associations concentrated in spawn and egg-availability windows, focal responses exceeded a prespecified specificity panel, and the strongest patterns recurred across regions. We linked complete eBird checklists to event-level Fisheries and Oceans Canada spawn records and used mixed models adjusted for effort, observer, year, location, and event structure. Analyses included 217,200 Strait of Georgia and 8,584 West Coast Vancouver Island checklist events. Near-spawn associations varied among taxa, guilds, outcomes, regions, and event windows. Surf Scoter and Short-billed Gull showed the clearest pattern: detection and conditional positive-count coefficients were positive in both regions. Event-window coefficients did not form a single resource-pulse trajectory. Matched spatial and observer sensitivities generally preserved direction, and shifted-exposure placebos were null after Benjamini-Hochberg adjustment. The Strait of Georgia specificity panel was non-null for Gadwall (beta = 0.234, 95% CI 0.132–0.337;  $q = 7.28 \times 10^{-6}$ ) and Northern Shoveler (beta = 0.628, 0.527–0.729;  $q = 6.56 \times 10^{-34}$ ). Pacific herring spawning therefore left a detectable but heterogeneous and not uniformly taxon-specific signature in eligible submitted checklists. The results support a partial and qualified resource-pulse interpretation, not a causal effect or population-level change.

**Keywords:** Pacific herring; eBird; community science; coastal birds; resource pulse; mixed models; falsification analysis